Tutorial 2.

1) Smart Attendance Checker.

```
import java.util.Scanner;

public class Smart_Attendance_Checker {
  public static void main (String [] args) {

      Scanner sc=new Scanner (System.in);

    System.out.print ("Enter total classes conducted:");
     int totalClasses =sc.nextInt();

    System.out.print ("Enter classes attended :");
     int attendedClasses= sc.nextInt();

    if (totalClasses ==0) {
      System.out.println(" Invalid data");
    }
    else if (attendedClasses > totalclasses ||
                attendedClasses < 0) {
      System.out.print (" Invalid Data");
    }
  else {
    double attendancePercentage = ((double)attendance-
                              Classes /totalClasses)* 100;
    System.out.printf("Attendance Percentage : %.2f%%\n",
                    attendance Percentage);

      if (attendancePercentage >=75) {
         System.out.println (" Eligible");
```

```
}else {
    System.out.println("Not Eligible");
  }
 }
 sc.close();
 }
}
```

2) AI Age Predictor

```
import java.util.Scanner;
import java.time.Year;

public class AI_Age_Predictor {
    public static void main (String[] args) {

        Scanner sc=new Scanner (System.in);

        System.out.print ("Enter your age :");
        int age= sc.nextInt();

        System.out.println("Age after 10 yrs :" +(age+10));
        System.out.println("Age after 25 yrs: "+(age+25));
        System.out.println("Age after 50 yrs:+(age+50));

        int currentYear=Year.now().getValue();
        int yearTurn100= currentYear +(100-age);

        System.out.println ("You will be 100 yrs in the
                      year:"+ yearTurn100);

        sc.close();
    }
}
```

3. Fitness Challenge Tracker

```
import java.util.Scanner;

public class Fitness_Challenge_Tracker {
    public static void (String[] args) {
        Scanner sc = new Scanner(System.in);

            int total = 0;
            int highest = 0;

        for (int i = 1; i <= 7; i++) {
            System.out.println("Enter steps for Day" + i + ":");
            int steps = sc.nextInt();

            total += steps;

            if (steps > highest) {
                highest = steps;
            }
        }
        double average = (double) total / 7;

        System.out.println("Total : " + total);
        System.out.println("Average : " + average);
        System.out.println("Highest : " + highest);

        sc.close();
    }
}
```

(Addition of 2 numbers)

4. Fibonacci Series 0+1=1+2=3+5=8

```
import java.util.Scanner;

  public class Fibonacci_Series {
    public static void main (String[] args) {
      Scanner sc = new Scanner (System.in);

        System.out.println ("Enter any number:");
          int n= sc.nextInt();

        int first=0 , second=1;

      System.out.println ("Fibonacci Series:");

      for (int i=1; i<=n; i++) {
        System.out.print (first + " ");

        int next= first + second;
          first = second;
          second= next;
      }
  sc.close();
   }
}
```

5. Prime Number Checker.

```
import java.util.Scanner;

 public class Prime_Number {
    public static void main (String[] args) {
      Scanner sc= new Scanner (System.in);

    System.out.print ("Enter any number:")
       int n= sc.nextInt();

       int count=0;

       if (n<0) {
       System.out.println ("Negative");
       }

       else if (n<=1) {
       System.out.println (" Not Prime");
       }

       else {
         for (int i=2; i<= Math.sqrt(n); i++) {
             if (n%i==0) {
                 count ++;
             }
       }
           if (count ==0) {
             System.out.println ("Prime");
          } else {
            System.out.println("Not Prime");
```

```
        }
    }
        sc.close();
    }
}
```

6.5. Logical Operators.

```
import java.util.Scanner;

public class Logical_Operators {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);

        int num1 = 0, num2 = 0, num3 = 0;

        for (int i = 1; i <= 3; i++) {
            System.out.println("Enter Number" + i + ":");

            if (i == 1) num1 = sc.nextInt();
            else if (i == 2) num2 = sc.nextInt();
            else if (i == 3) num3 = sc.nextInt();
        }
        boolean isSum = (num3 == (num2 + num1));

        if (isSum) {
            System.out.println(num3 + "is sum of" + num1 "and"
                                + num2);
        } else {
            System.out.println( num3 + "is not sum of" + num1 "and"
                                + num2 );
        }
    sc.close(); }}
```

7 Multiplication

```
import java.util.Scanner;

   public class Multiplication {
      public static void main(String[] args) {
         Scanner sc=new Scanner(System.in);

         System.out.print("~~Multiplication~~ Enter any
                                 number:");
         int num=sc.nextInt();

         for (int i=1; i<=10; i++) {
            System.out.println("Multiplication table"+ num +
                          "*" +i + "="+(num * i));
         }

         sc.close();
      }
   }
```

8. Right Angled Triangle Pattern

```
import java.util.Scanner;

public class Right_Angled_Triangle {
  public static void main (String [] args) {
    Scanner sc = new Scanner (System.in);

    System.out.println ("Enter any number of rows : ");
      int rows = sc.nextInt();

    for (int i = 1; i <= rows; i++) {

      for (int j = 1; j <= i; j++) {
        System.out.print (" * ");
      }

      System.out.println (" ");
    }
  sc.close();
  }
}
```

9. Pyramid Pattern

```
import java.util.Scanner;

    public class Pyramid-Pattern {
       public static void main (String [] args) {
       Scanner sc=new Scanner (System.in);

       System.out.print ("Enter number of rows :");
          int rows= ~~sc~~next Int();

       for (int i=1; i<=rows; i++) {
        ~~{~~
          for (int j=i; j<=rows; j++) {
             System.out.print (" ");
          }

        for (int k=1; K<=(2*i-1); k++) {
            System.out.print ("*");
        }

        ~~System~~ System.out.println();
         sc.close();
        }
     }
  }
```

10. Palindrome Number (A number that reads the same forward and backward)

121 → 121 ✓

321 → 123 ✗

```
import java.util.Scanner;

public class Palindrome_Number {
    public static void main (String [] args) {
        Scanner sc=new Scanner (System.in);

        System.out.print ("Enter any number : ");
        int num=~~next Int();~~ sc.nextInt();

        int org_num=num;
        int rev = 0;

        while (num!=0) {
            rev= rev * 10 + num%10;
            num= num /10
        }
        if (org_num==rev) {
            System.out.println (" org_num +"is Palindrome");
        } else {
            System.out.println(org_num+" is not Palindrome");
        }
        sc.close();
    }
}
```

11. Basic Calculator.

```
import java.util.Scanner();

 public class Calculator {
    public static void main(String[] args) {
       Scanner sc=new Scanner(System.in);

    System.out.println("Enter first number:");
       double a= sc.nextDouble();

    System.out.println("Enter second number:");
       double b= sc.nextDouble();

    System.out.println("Enter operator :");
       char op= sc.next().charAt(0);

    switch(op) {

    case '+' :System.out.println(a + b); break;
    case '-' : System.out.println(a-b); break;
    case '*' :System.out.println(a *b); break;
    case '/':
       if(b!=0) System.out.println(a/b);
    else System.out.println("Error");
       break;
    default :System.out.println("Invalid operator");
    }
 sc.close();
 }
}
```

12. Grade Calculator

```
import java.util.Scanner;

  public class Calculator {
    public static void main(String[] args){
    Scanner sc=new Scanner(System.in);

  System.out.println("Enter marks obtained:");
    double marks = sc.nextDouble();

  String grade=

      (marks < 0 || marks >100) ? "Invalid Marks":
      (marks >=90) ? "A+" :
      (marks>=80) ? "A":
      (marks >=76)? "A-":
      (marks >= 72)? "B+":
      (marks>=68) ? "B":
      (marks>=65)? "B-":
      (marks>=60) ? "C+":
      (marks>= 56)? "C":
      (marks>=50)? "C-":
      (marks >=40) ? "D":
      "F";

    System.out.println("Grade =" +grade);
  sc.close();
  }
}
```